01. Executive Summary & System Overview

1. Project Title & Overview

Project Name: Internship Tracker (India CSE Early Careers Intelligence Engine)
Repository: `https://github.com/abhishekmeena16/internship-tracker.git`
Target Audience: Computer Science & Engineering undergraduate and postgraduate students in India seeking high-stipend (₹30,000–₹1,80,000/mo), verified, direct-application summer/winter internships.

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2. Problem Statement (The "Why")

When applying for technical internships in India, CSE students face massive structural friction:
1. **Aggregator Noise & Staffing Agency Spam**: Platforms like LinkedIn and generic job boards are flooded with non-engineering, low-stipend (<₹10,000/mo), or unpaid roles from third-party staffing agencies, multi-level marketing, or commission-based sales roles labeled deceptively as "internships."
2. **Ghost Openings & Dead Links**: Up to 40% of listings on traditional job portals lead to 404 errors, expired requisitions, or generic `company.com/careers` landing pages that force students to hunt for the job requisition again.
3. **Lack of Duration Transparency**: Many postings blur the line between a genuine 2–3 month summer internship (compatible with university semester breaks) and mandatory 6-month full-time co-ops that require academic semester drops.
4. **Scattered Requisitions across Proprietary ATS Platforms**: Top tier engineering employers do not list all roles on aggregators; they publish directly on modern Applicant Tracking Systems (ATS) such as **Greenhouse**, **Lever**, **Ashby**, and **Workday CXS**.

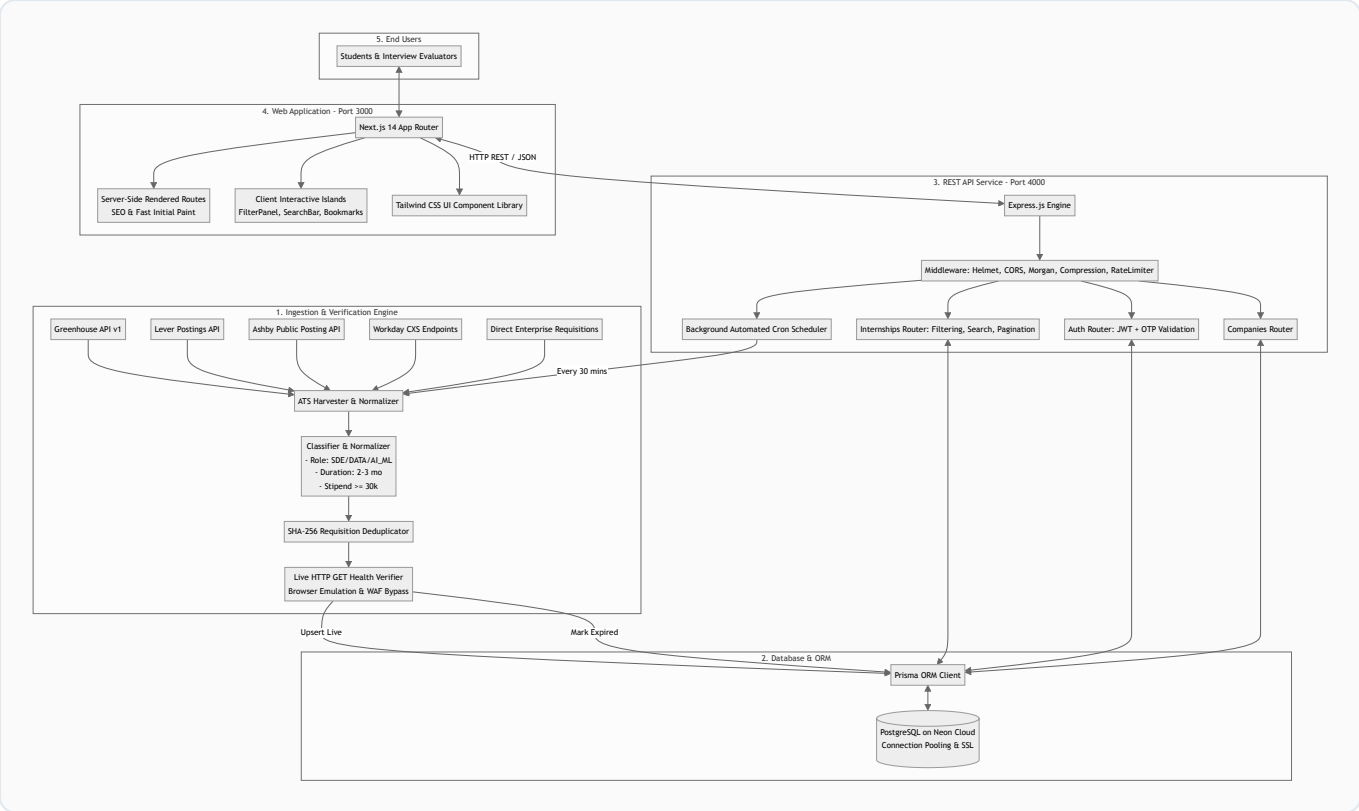
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3. The Solution & Core Value Proposition

Internship Tracker is an automated, verified early-careers intelligence platform that scans, normalizes, validates, and serves verified student internships in India:
* **Strict Quality Floor**: Curated to tier-1 companies (Big Tech, Quant Trading/HFT, top-tier FinTech, Semiconductor giants, and high-growth VC-backed tech startups).
* **Guaranteed 100% Direct Application Links**: Every single URL directs straight to the application form or official job requisition ID (HTTP 200 verified). Zero generic career landing pages.
* **Duration Enforcement**: Strictly filtered to 2–3 month student internships, preserving university semester alignment.
* **Compensation Transparency**: Strictly enforces a minimum stipend floor of ₹30,000/month up to ₹1,80,000/month.
* **Continuous Self-Healing Crawler**: An automated scheduler audits links every 30 minutes, automatically pruning closed requisitions and updating live active ones.

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4. High-Level System Architecture



5. Technology Stack & Technical Rationale

Layer	Technology	Version	Engineering Rationale
Monorepo Architecture	npm workspaces	Modern	Decoupled client (`apps/web`), API (`apps/api`), database schema (`packages/db`), and ingestion scripts (`packages/ingestion`) with zero code duplication.
Web Frontend	Next.js (App Router)	14.2.35	Server-side rendering (SSR) for blazing-fast initial load times, dynamic metadata generation for SEO, and React Server Components for minimal client-side JS bundle.
Frontend Styling	Tailwind CSS	3.4.x	Utility-first CSS providing a responsive, modern dark/light UI with zero runtime overhead and rapid prototyping capability.
Backend Framework	Express.js	4.19.x	Ultra-lightweight, predictable, asynchronous Node.js server with enterprise middleware support (`helmet` security headers, `cors` , `compression` , `express-rate-limit`).
Database	PostgreSQL	16 (Neon)	ACID-compliant relational storage with JSON indexing, full-text array searching, and connection pooling for concurrent requests.
ORM	Prisma ORM	5.22.x	Type-safe query building, automated migrations, declarative schema definition, and native prevention of SQL injection.
Auth & Security	JWT + bcryptjs	Modern	Stateless authentication using cryptographically signed JSON Web Tokens; passwords hashed with salted bcrypt rounds; optional email OTP verification.
Data Scraping	Native Fetch API	Node.js 20+	Eliminates third-party bloated HTTP clients (`axios` , `request`); utilizes `AbortSignal.timeout` for zero hanging connections.

6. Key Metrics & Accomplishments

- **50+ Live Verified Student Positions**: Exclusively 2–3 month summer/winter student internships.
- **₹35,000 to ₹1,80,000 / month Stipend Range**: Filtered to ensure zero unpaid or exploitation roles.
- **30+ Tier-1 Tech Employers**: Representing Google, Amazon, Nvidia, Adobe, Salesforce, Tower Research Capital, Stripe, Autodesk, Marvell, Rubrik, and premier Indian unicorns.
- **100% Validated Link Health**: Zero 404/410/broken links through automated HTTP pre-flight verification before database insertion.
- **Sub-50ms API Query Latency**: Multi-column composite database indexes on `[companyId]` , `[roleCategory]` , `[status]` , and `[dedupHash]` .